

CRAMÉR–VON MISES TYPE TESTS FOR SYMMETRY

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SUMMARY

A new class of quadratic rank statistics is introduced which are used to test the null hypothesis of symmetry of a distribution function about zero. The class is created by the insertion of an arbitrary weight function which parallels a well-known development for Cramér–Von Mises goodness-of-fit statistics. Monte Carlo power comparisons for sample size 20 are made for three tests based on members of this new class and the Wilcoxon signed rank test. Limiting distributions for the quadratic rank statistics are discussed. For testing against alternative sequences of distribution functions converging to a symmetric distribution function, asymptotically optimal tests are obtained which give the greatest increase in the asymptotic mean under the alternatives in units of the asymptotic null hypothesis standard deviation.

1. Introduction

Given a sample $X_1, \dots, X_n$ from a continuous cdf F , consider testing the null hypothesis H_0 : F is symmetric about zero. We study a class of test statistics which is generated from a Cramér–Von Mises type statistic for symmetry discussed by Rothman and Woodroffe (1972). The class is generated by the insertion of an arbitrary weight function that parallels a similar treatment for the Cramér–Von Mises goodness-of-fit statistic (see Anderson and Darling (1952) and Gregory (1977)).

Let F_n be the sample distribution function and $F'_n(x) = [F_n(x + 0) + F_n(x - 0)]/2$ for $x \in \mathbb{R}$. Consider statistics

$$(1.1) \quad R_n = n \int_{-\infty}^{\infty} [F_n(x) + F'_n(-x) - 1]^2 w[F'_n(x)] dF_n(x),$$

where $w(\cdot)$ is a non-negative weight function on $(0,1)$. If w is symmetric about $1/2$, the use of F'_n instead of F_n makes R_n invariant under multiplication of the data by -1 . Except for Theorem 3.2 we assume w to be symmetric about $1/2$. The statistic may be rewritten in terms of ranks. If $X_{(1)} \leq \dots \leq X_{(n)}$ is the ordered sample and R_i^+ is the rank of $|X_{(i)}|$ in $\{|X_{(1)}|, \dots, |X_{(n)}|\}$, then

$$R_n = n^{-2} \sum_{i, X_{(i)} \leq 0} [R_i^+ - (n - 2(i - 1) - 1/2)]^2 w[(2i - 1)/2n] \\ + n^{-2} \sum_{i, X_{(i)} > 0} [R_i^+ - (n - 2(n - i) - 1/2)]^2 w[(2i - 1)/2n].$$

The test rejects for large values of R_n . The statistic of Rothman and Woodroffe (1972) corresponds to $w \equiv 1$. In Section 2 we consider the tests based on weight functions $w \equiv 1$, $w[(1 + x)/2] = 1/2 + |x|$ and $w[(1 + x)/2] = (1 - x)^{-1} (1 + x)^{-1}$. Monte Carlo power results are given for these three quadratic rank tests as well as for the locally most powerful linear rank test against logistic location alternatives, the Wilcoxon signed rank test (see Hájek and Sidák (1967)). The asymptotic results of Section 3 show that the test based on $w[(1 + x)/2] = 1/2 + |x|$ satisfies an optimum property in the class R_n against logistic location alternatives. The four tests considered in Section 2 are found to have similar power properties for logistic location alternatives. The test based on $w[(1 + x)/2] = (1 - x)^{-1} (1 + x)^{-1}$ is shown to have superior power for some specifically constructed alternatives that are approximately symmetric near the median (zero) but which differ dramatically in the tails. Some percentage points for the quadratic rank statistics are given in Section 2.

Section 4 discusses the connection with tests for uniformity.

Statistics similar to the case $w \equiv 1$ are discussed in Orlov (1972) and Srinivasan and Godio (1974).

2. Monte Carlo results

Let W be the Wilcoxon signed rank statistic, $W = \sum_{i, X_{(i)} > 0} R_i^+$. Let

$$RW = \{\text{expression (1.1)}\} \text{ with } w \equiv 1,$$

$$G1 = (1, 12) \{\text{expression (1.1)}\} \text{ with } w[(1 + x)/2] = 1/2 + |x|,$$

and

$$G2 = (0, 62) \{\text{expression (1.1)}\} \text{ with } w[(1 + x)/2] = (1 - x)^{-1} (1 + x)^{-1}.$$

The multipliers are introduced to bring the upper percentage points of the three quadratic rank statistics into the same range for tabulation convenience (see Table 2 below).

Table 1 presents the results of a power study involving tests based on these four statistics. Two-tailed tests based on (1.1) reject for large values of the statistic. One-tailed tests for positive or negative shifts in location based on (1.1), may be constructed by using (1.1) (an even function of the observations) in conjunction with an odd function of the observations (sample mean or median) in the obvious way. The one-tailed tests based on (1.1) of Table 1 reject for large values of (1.1) if the sample mean is positive. The

terminology logistic + d refers to the density $f(x - d)$

where $f(x) = e^x(1 + e^x)^{-2}$, $-\infty < x < \infty$, while $\chi^2(4) + d$ refers to the density $f(x - d)$ where $f(x) = xe^{-x/2}/4$, $x > 0$. The densities f_1 and f_3 were constructed to have very different tails but otherwise to be fairly symmetric. The form of these densities is

$$f_k(x) = \begin{cases} (x+1)^{-1+1/k/2k}, & -1 < x < 0 \\ (1+x/k)^{-2/2k}, & 0 < x \end{cases}$$

TABLE 1

Power of 5% test with 20 samples (1 000 simulations)

One-tailed tests for positive mean				Distribution	Two-tailed tests			
W	RW	G1	G2		W	RW	G1	G2
0,046	0,052	0,048	0,052	H_0 (symmetry)	0,053	0,057	0,055	0,052
0,122	0,134	0,132	0,128	logistic +0,2	0,073	0,080	0,078	0,067
0,251	0,264	0,259	0,259	logistic +0,4	0,160	0,169	0,163	0,158
0,437	0,448	0,442	0,440	logistic +0,6	0,316	0,334	0,321	0,314
0,629	0,642	0,638	0,623	logistic +0,8	0,493	0,501	0,494	0,496
0,117	0,111	0,122	0,191	$\chi^2(3) - 3,36$	0,072	0,071	0,076	0,112
0,007	0,009	0,011	0,021	$\chi^2(4) - 4$	0,115	0,150	0,137	0,115
0,176	0,193	0,218	0,350	f_1 (see text)	0,105	0,105	0,111	0,209
0,309	0,381	0,411	0,704	f_3 (see text)	0,197	0,223	0,251	0,512

The upper 10% and 5% points used in Table 1 were determined from 5 000 simulations as follows: RW: 1,1425 and 1,5875; G1: 1,1290 and 1,5954 and G2: 1,0900 and 1,5698.

We make three points concerning Table 1 and comment on the validity of the results:

- 1) The quadratic rank test designed for logistic location alternatives, G1, is dominated by RW for these alternatives but the difference is small. Indeed, all three quadratic rank tests performed essentially the same for these alternatives. This is in accord with the closeness of the optimality criteria (the right-hand side of (3.7)) for quadratic rank statistics as discussed in Section 3.

TABLE 2
Null Hypothesis Cumulative Distribution Function Values at $x = 1 (0,25)3$ for
RW, G1 and G2 (5 000 simulations)

	$x = 1,0$			$x = 1,25$			$x = 1,5$			$x = 1,75$			$x = 2,0$		
n	RW	G1	G2	RW	G1	G2	RW	G1	G2	RW	G1	G2	RW	G1	G2
5	0,8764	0,8764	0,8764	0,8764	0,8764	0,8764	0,9362	0,9362	0,9362	1,0	0,9362	0,9362	1,0	1,0	1,0
10	0,8772	0,8754	0,8744	0,9144	0,9088	0,9094	0,9416	0,9390	0,9368	0,9692	0,9568	0,9578	0,9796	0,9740	0,9674
15	0,8654	0,8714	0,8778	0,9094	0,9112	0,9160	0,9406	0,9410	0,9432	0,9618	0,9608	0,9614	0,9744	0,9722	0,9730
20	0,8686	0,8722	0,8840	0,9150	0,9218	0,9248	0,9432	0,9466	0,9500	0,9610	0,9616	0,9652	0,9732	0,9732	0,9756
25	0,8672	0,8734	0,8846	0,9070	0,9134	0,9222	0,9340	0,9390	0,9476	0,9572	0,9586	0,9654	0,9706	0,9728	0,9794
30	0,8604	0,8706	0,8880	0,9076	0,9162	0,9236	0,9358	0,9408	0,9464	0,9566	0,9590	0,9624	0,9690	0,9706	0,9780
40	0,8638	0,8758	0,8898	0,9106	0,9194	0,9302	0,9368	0,9458	0,9556	0,9582	0,9644	0,9716	0,9730	0,9770	0,9916
50	0,8666	0,8802	0,8956	0,9076	0,9164	0,9294	0,9348	0,9414	0,9544	0,9554	0,9630	0,9690	0,9698	0,9752	0,9802
∞	0,8639			0,9080			0,9370			0,9564			0,9696		

	$x = 2,25$			$x = 2,5$			$x = 2,75$			$x = 3,0$		
n	RW	G1	G2	RW	G1	G2	RW	G1	G2	RW	G1	G2
5	1,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0
10	0,9868	0,9812	0,9786	0,9908	0,9868	0,9818	0,9936	0,9908	0,9872	0,9956	0,9922	0,9882
15	0,9834	0,9804	0,9812	0,9896	0,9878	0,9872	0,9934	0,9916	0,9910	0,9952	0,9940	0,9928
20	0,9818	0,9816	0,9832	0,9878	0,9868	0,9860	0,9908	0,9904	0,9896	0,9934	0,9932	0,9914
25	0,9840	0,9842	0,9854	0,9902	0,9898	0,9898	0,9936	0,9928	0,9936	0,9962	0,9960	0,9962
30	0,9810	0,9820	0,9842	0,9878	0,9884	0,9912	0,9926	0,9924	0,9926	0,9944	0,9944	0,9952
40	0,9818	0,9840	0,9870	0,9884	0,9896	0,9916	0,9932	0,9932	0,9940	0,9948	0,9944	0,9954
50	0,9798	0,9826	0,9866	0,9860	0,9888	0,9896	0,9908	0,9918	0,9934	0,9930	0,9940	0,9958
∞	0,9787			0,9850			0,9894			0,9925		

- 2) The test based on G2 does best where it was anticipated that it would do so (densities f_1 and f_3).
- 3) For logistic location alternatives the quadratic rank test based on RW does at least as well as the locally most powerful linear rank test (W). Some suspicion is aroused by the dominance of G1 by RW for location alternatives. This would be the result if the percentage points of RW were too low or those of G1 were too high. Some experimentation with different percentage points in a range consistent with the estimates based on 5 000 simulations, failed to change the conclusions 1), 2), and 3).

Table 2 presents null hypothesis cumulative distribution function values for the three quadratic rank statistics for a selection of sample sizes. The values for RW when $n = \infty$ were calculated from the asymptotic expansion in Rothman and Woodroffe (1972). The convergence of this expansion was extremely rapid for values x in the range we considered (2 or 3 terms will suffice). The rapidity of this convergence was not noted in Rothman and Woodroffe (1972).

Finally, we note the rapid convergence to the asymptotic values in Table 2. This seems characteristic of Cramér–Von Mises type statistics (see e.g., Stephens and Maag (1968) in connection with the one sample Cramér–Von Mises test of fit).

3. Limiting distributions and asymptotically optimal tests

In this section the limiting distribution of R_n is discussed and manageable expressions are obtained for the mean and variance of the limiting distribution under alternatives. An optimum weight function is found for logistic location alternatives.

The limiting distribution of R_n is obtained on ' or weight functions w satisfying

Assumption 3.1

$w(u)$, $0 \leq u \leq 1$, is continuous with a bounded derivative existing except for at most a finite number of points in $[0, 1]$.

Let $Q_n(x) = n^{1/2} [F_n(x) + F_n(-x) - 1]$, $x \in \mathbb{R}$, and Q'_n be Q_n with F_n replaced by F'_n . It follows that

$$R_n = \int_{-\infty}^{\infty} [Q'_n(x)]^2 w[F'_n(x)] dF_n(x).$$

Define

$$R_{n0} = \int_{-\infty}^{\infty} [Q'_n(x)]^2 w[F(x)] dF_n(x),$$

$$R_{n1} = \int_{-\infty}^{\infty} [Q_n(x)]^2 w[F(x)] dF_n(x),$$

and

$$(3.1) R_{n2} = \int_{-\infty}^{\infty} [Q_n(x)]^2 w[F(x)] dF(x).$$

We will show $R_n - R_{n2} \xrightarrow{P} 0$ under H_0 by showing $R_n - R_{n0} \xrightarrow{P} 0$, $R_{n0} - R_{n1} \xrightarrow{P} 0$, and $R_{n1} - R_{n2} \xrightarrow{P} 0$ under H_0 .

Theorem 3.2

Suppose w satisfies Assumption 3.1. If $X_1, \dots, X_n$ is a sample from a continuous symmetric distribution F then $R_n - R_{n2} \rightarrow 0$ in probability as $n \rightarrow \infty$.

Proof:

Using

$$(1) |F_n(x) - F'_n(x)| \leq 1/2n, x \in R,$$

$$(2) |w(u) - w(v)| \leq B|u - v|, 0 < u, v < 1, \text{ for some } B \in (0, \infty) \text{ and}$$

$$(3) n^{1/2} \sup_{x \in R} |F_n(x) - F(x)| = O_p(1)$$

we easily obtain $R_n - R_{n0} \xrightarrow{P} 0$ and $R_{n0} - R_{n1} \xrightarrow{P} 0$. The last approximation $R_{n1} - R_{n2} \xrightarrow{P} 0$, is treated by a technique in the proof of Theorem 2 of Rothman and Woodroffe (1972). Referring to that work, we see that for any integer k ,

$$(3.2) |R_{n1} - R_{n2}| \leq 2 \sup_{|s-t| \leq 1/k} |[V_n(t)]^2 - [V_n(s)]^2| \\ + 2 \left(\sup_t [V_n(t)]^2 \right) \left(\sum_{j=0}^k |H_n(j/k) - j/k| \right),$$

where $V_n(t) = Q_n(F^{-1}(t)) w^{1/2}(t)$ and $H_n(t) = F_n(F^{-1}(t))$, $0 < t < 1$, with $F^{-1}(t) = \inf \{x: F(x) \geq t\}$. The continuity of w on $[0,1]$ together with standard convergence results related to the process $Q_n(F^{-1}(t))$, $0 < t < 1$, yield $R_{n1} - R_{n2} \xrightarrow{P} 0$ by letting $n \rightarrow \infty$ and then $k \rightarrow \infty$ in (3.2). //

We have immediately that tests which reject for large values of R_n are consistent against all non-symmetric alternatives if w is positive on $(0,1)$. By Theorem 3.2 large sample distribution theory for R_n under H_0 is the same as that for R_{n2} (3.1). Now since R_n is distribution free under H_0 we take without loss of generality

$$F(x) = F_0(x) = \begin{cases} 0 & x < -1 \\ (x+1)/2 & -1 \leq x \leq 1 \\ 1 & 1 < x \end{cases}$$

Now one can write

$$R_{n2} = n^{-1} \sum_{i,j} Q(X_i, X_j),$$

where

$$(3.3) \quad Q(s, t) = \int_{-1}^1 \{I[x \leq s] - I[s \leq -x]\} \{I[x \leq t] - I[t \leq -x]\} w[(1+x)/2] d(x/2),$$

$$-1 \leq s, t \leq 1.$$

The results of Gregory (1977) apply to give the limiting distribution of R_{n2} as that of $\sum_k \lambda_k Z_k^2$ where $\{\lambda_k\}$ are the eigenvalues of Q and $Z_1, Z_2, \dots$ are i.i.d. standard normal variables.

Theorem 3.3

Suppose that w satisfies

- (1) w is symmetric about $1/2$
- (2) w' exists almost everywhere on $(0, 1)$
- (3) w is bounded on $[\epsilon, 1 - \epsilon]$ for each $\epsilon > 0$
- (4) Q is square integrable.

Then $g(s)$, $-1 < s < 1$, is an eigenfunction of Q (3.3) corresponding to a non-zero eigenvalue λ if and only if

- (i) g is odd and absolutely continuous,
- (ii) $g'(s)/w[(1+s)/2]$, $-1 < s < 1$, is absolutely continuous and approaches zero as $s \uparrow 1$,
- (iii) $\lambda[g'(s)/w[(1+s)/2]]' + g(s) = 0$ for almost all s in $(-1, 1)$.

Proof:

If g is an eigenfunction of Q corresponding to the eigenvalue λ , then

$$(3.4) \quad \int_{-1}^1 Q(s, t) g(t) dF_0(t) = \lambda g(s), \text{ a.e. (Lebesgue measure) } [-1, 1].$$

Substitution of (3.3) and interchanging the order of integration yields

$$(3.5) \quad (\text{sign } s) \int_{|s|}^{|s|} \left\{ - \int_{-1}^{-|x|} g + \int_{|x|}^1 g \right\} w[(1+x)/2] d(x/2) = 2\lambda g(s), \quad -1 < s < 1.$$

Interchanging the order of integration follows from Fubini's theorem by using condition

(3) of the theorem and the integrability of g . Differentiating (3.5) and using the symmetry of w yields

$$(3.6) \quad \left\{ -\int_{-1}^{-|s|} g + \int_{|s|}^1 g \right\} w[(1+s)/2] = 2\lambda g'(s), \quad -1 < s < 1.$$

Now part (iii) of the theorem follows and the other parts follow by using (3.5) and (3.6). We may reverse the sequence of steps to show that any g and $\lambda (\neq 0)$ satisfying parts (i), (ii), and (iii) of the theorem also satisfy (3.4). //

When $w \equiv 1$ the eigenvalues $\lambda_0 = 0, \lambda_k = 4\pi^{-2}(2k-1)^{-2}, k = 1, 2, \dots$ are readily obtained and of course are the ones found in Rothman and Woodroffe (1972) by another method. For this case ($w \equiv 1$) the differential equation appearing in Theorem 3.3 is the same as one obtained in MacNeill (1974) which relates to the limiting distribution of a class of statistics different from R_n . In this regard see also De Wet and Venter (1976) and in particular Section 3 where an equation similar to our equation (iii) of Theorem 3.3 is discussed but with different boundary conditions. Finally, in Section 4 we relate the eigenvalue problem here to the one considered in De Wet and Venter (1973).

The results of Gregory (1977) allow us to obtain the limiting distribution of R_n under an alternative sequence converging to the null hypothesis. Suppose $X_1, \dots, X_n$ are i.i.d. with common df F_{0n} where F_{0n} converges to F_0 as $n \rightarrow \infty$. We calculate the efficacy described in Gregory (1977) which is the increase in the asymptotic mean under the alternatives in asymptotic null hypothesis standard deviation units. Let F_{0n} have a Radon-Nikodym derivative with respect to F_0 , $dF_{0n}/dF_0 = 1 + n^{-1/2} h_n$, where h_n converges to a function h in the space of square integrable (with respect to Lebesgue measure) functions on $[-1, 1]$. Following Gregory (1977) define the efficacy

$$e_{w,h} = \left\{ \int_{-1}^1 \int_{-1}^1 Q(s,t) h(s) h(t) d(s/2) d(t/2) \right\} \left\{ 2 \int_{-1}^1 \int_{-1}^1 Q^2(s,t) d(s/2) d(t/2) \right\}^{-1/2}$$

Careful calculation shows that

$$(3.7) \quad e_{w,h} \sim \frac{\int_0^1 \left\{ \int_{-1}^{-x} h - \int_x^1 h \right\}^2 w[(1+x)/2] dx}{\left\{ \int_0^1 \int_0^1 \{ \min(1-x, 1-y) \}^2 w[(1+x)/2] w[(1+y)/2] dx dy \right\}^{1/2}}$$

The following lemma can be used to obtain an optimum choice for w given a fixed alternative sequence generated by h .

Lemma 3.4

Let H and ξ have continuous derivatives on $(0,1)$. If

$$\lim_{x \rightarrow 1} H^2(x)/\xi(x) = \lim_{x \rightarrow 1} (1-x)\xi(x) = 0 \text{ and } \lim_{x \rightarrow 0} H^2(x)/\xi(x) = \lim_{x \rightarrow 0} \xi(x) = 0$$

then

$$\begin{aligned} & \frac{\int_0^1 H^2(x) d\xi(x)}{\left\{ \int_0^1 \int_0^1 \{ \min(1-x, 1-y) \}^2 d\xi(x) d\xi(y) \right\}^{1/2}} \\ &= \frac{2^{1/2} \int_0^1 [\xi(x)/(1-x)^{1/2}] [-H(x)H'(x)/(1-x)^{-1/2}] dx}{\left\{ \int_0^1 [\xi(x)/(1-x)^{1/2}]^2 dx \right\}^{1/2}} \end{aligned}$$

The proof is omitted.

Comparison with (3.7) and using the Cauchy-Schwarz inequality shows that the optimum w satisfies

$$(3.8) \quad w[(1+x)/2] = -[H(x)H'(x)(1-x)^{-1}]',$$

where

$$H(x) = \int_{-1}^{-x} h - \int_x^1 h, \quad 0 < x < 1.$$

Consider logistic location alternatives. After making the transformation to the interval $[-1, 1]$ we have

$$F_{0n}(x) = F \{ F^{-1} [(1+x)/2] + \delta n^{-1/2} \}, \quad -1 < x < 1,$$

where

$$F(x) = (1 + e^{-x})^{-1}, \quad -\infty < x < \infty.$$

It is easy to show that the required function h is

$$h(x) = \delta f' [F^{-1} [(1+x)/2]] / f [F^{-1} [(1+x)/2]], \quad -1 < x < 1$$

where f is the logistic density. Now it follows (take $\delta = 1$) that $h(x) = -x$, $-1 < x < 1$, and $H(x) = 1 - x^2$, $0 < x < 1$. Substitution into (3.8) gives $w[(1+x)/2] = 1/2 + x$,

$0 < x < 1$. By symmetry, the weight function w which maximizes the efficacy (3.7) for logistic location alternatives is

$$w[(1+x)/2] = 1/2 + |x|, \quad -1 < x < 1.$$

This weight function w and the function $h(x) = -x$ for logistic location alternatives gives the value 1,31656 for the right-hand side of (3.7). This is only slightly above the values 1,31087 for $w[(1+x)/2] = (1+x)^{-1} (1-x)^{-1}$ and 1,30639 for $w \equiv 1$. The closeness of the efficacies is in accord with the closeness of the powers found in Section 2.

4. The relation with the Cramér-Von Mises test for uniformity

We have shown that when $F = F_0$, the uniform cdf on $[-1,1]$, the statistic R_n is asymptotically equivalent to

$$\begin{aligned} (4.1) \quad R_{n2} &= n \int_{-1}^1 [F_n(x) + F_n(-x) - 1]^2 w[(1+x)/2] d[(1+x)/2] \\ &= n^{-1} \sum_{i,j} Q(X_i, X_j), \end{aligned}$$

where Q is given by (3.3). Now when F is a cdf on $[-1,1]$ which is symmetric about zero it is found that

$$\begin{aligned} (4.2) \quad n \int_{-1}^1 [F_n(x) - F(x)]^2 w[(1+x)/2] d[(1+x)/2] \\ = R_{n2}/4 + \int_{-1}^1 [F_n^*(x) - F(x)]^2 w[(1+x)/2] d[(1+x)/2] \end{aligned}$$

where

$$F_n^*(x) = [F_n(x) + 1 - F_n(-x)]/2.$$

We have F_n^* as the natural estimate of $F(x)$ under the symmetry assumption. From (4.2) it is seen that F_n^* may be described as a least squares estimate of $F(x)$.

Putting F equal to the uniform cdf in (4.2) gives an equation whose left-hand side is the Cramér-Von Mises statistic for uniformity. The total deviation from uniformity (left-hand side of (4.2)) may then be said to be the sum of two components, the deviation due to lack of symmetry ($R_{n2}/4$) and the deviation from uniformity given symmetry. Writing the quantities in (4.2) in terms of kernels as in (4.1) we have

$$(4.3) \quad n^{-1} \sum_{i,j} C^*(X_i, X_j) = 1/4 [n^{-1} \sum_{i,j} Q(X_i, X_j)] + 1/4 [n^{-1} \sum_{i,j} P(X_i, X_j)]$$

where

$$C^*(s,t) = \int_{-1}^1 \{I[s \leq x] - (1+x)/2\} \{I[t \leq x] - (1+x)/2\} \\ w[(1+x)/2] d[(1+x)/2],$$

and

$$P(s,t) = \int_{-1}^1 \{I[x \leq s] + I[s \leq -x] - 1 + x\} \{I[x \leq t] + I[t \leq -x] - 1 + x\} \\ w[(1+x)/2] d[(1+x)/2], -1 < s, t < 1.$$

The corresponding relation between kernels is

$$(4.4) \quad C^*(s,t) = 1/4 Q(s,t) + 1/4 P(s,t).$$

The kernel C^* for testing uniformity on $[-1,1]$ is related to the kernel C_w of De Wet and Venter (1973) for testing uniformity on $[0,1]$ by

$$(4.5) \quad C^*(s,t) = C_w((1+s)/2, (1+t)/2), -1 < s, t < 1.$$

The eigenvalues of C^* are the same as those of C_w and eigenfunctions of C^* may be obtained from eigenfunctions g of C_w as $g[(1+s)/2]$, $-1 < s < 1$.

It is seen that Q is odd in its arguments and P is even, i.e.

$$Q(s,t) = -Q(-s,t) \text{ and } P(s,t) = P(-s,t).$$

Then the eigenfunctions of Q must be odd (as noted) and the eigenfunctions of P must be even. Moreover it is easily shown that if g is odd

$$\int_{-1}^1 P(s,t)g(t) d(t/2) = 0$$

and that if g is even

$$\int_{-1}^1 Q(s,t)g(t) d(t/2) = 0.$$

Using (4.4) we conclude that g, λ is an eigenpair for $Q(P)$ if and only if $g, \lambda/4$ with g odd

(even) is an eigenpair for C^* . Therefore, the eigenvalues of Q may be obtained simply from those of C^* which in turn may be obtained from those of C_w (4.5) as discussed by De Wet and Venter (1973). The eigenvalues corresponding to

$$w[(1+x)/2] = (1+x)^{-1} (1-x)^{-1}$$

may be so obtained. Theorem 3.2 does not cover this case but one expects the result to hold in this case also.

We conclude by commenting on the similarity between (4.3) and the analysis of variance technique of partitioning a total sum of squares into independently distributed chi-square components. A heuristic development is as follows. Write

$$Q(s,t) = \sum \lambda_K g_K(s) g_K(t)$$

and

$$P(s,t) = \sum \gamma_K f_K(s) f_K(t),$$

where $\{\lambda_K, g_K\}$ ($\{\gamma_K, f_K\}$) are the eigenpairs for $Q(P)$. Then the right-hand side of (4.3) is

$$1/4 \sum \lambda_K \left[n^{-1/2} \sum_{i=1}^n g_K(X_i) \right]^2 + 1/4 \sum \gamma_K \left[n^{-1/2} \sum_{i=1}^n f_K(X_i) \right]^2.$$

The terms in brackets above approach independent standard normal variables in distribution.

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